MTH108 2022S1 Exam

Feel free to contact me at maths_and_programming@outlook.sg if you need tuition. I provide university level tuition for the following SUSS units:

* ENG301 Microprocessor Programming
* ENG355 Machine Learning
* ICT133 Structured Programming
* ICT162 Object Oriented Programming
* MTD215 Application of C++ in Multimedia
* ICT231 Data Structures and Algorithms I
* ICT232 Data Structures and Algorithms II
* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH319 Mathematical Methods I
* MTH320 Mathematical Methods II
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference

Donaldson Tan

MTH108

Timed Online Assignment – January Semester 2022

Calculus II

Tuesday, 24 May 2022 **4:00 pm – 6:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This Timed Online Assignment (TOA) contains **FIVE (5)** questions and comprises **FIVE (5)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O)*.

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

Answer all questions. (Total 100 marks)

Question 1

Use various techniques of integration to evaluate the following integrals:

(a) $\int (\sin x)^{2022} \cos x \, dx$.

(6 marks)

(b) $\int 3x \cos(1 - 2x) \, dx$.

(7 marks)

(c) Determine the value of $\int_1^2 \frac{x+3}{x^3+3x^2+2x} dx$.

(7 marks)

Question 2

Let R be the region bounded by the curve $y = -x^2 - 3x + 10$ and the line $y = 5 + x$.

Apply integration to solve the following problems.

(a) Find the area of the region R.

(6 marks)

(b) Using the disk/washer method, find the exact volume of the solid of revolution obtained when R is rotated 2π radians about the x-axis.

(7 marks)

(c) Using the method of cylindrical shells, find the exact volume of the solid of revolution obtained when R is rotated 2π radians about the line $x = 3$.

(7 marks)

Question 3

(a) Solve $\frac{d}{dx}\int_{e^x}^{x^2} \cos(\cos t)\, dt$.

(6 marks)

(b) Determine the derivative $f'(x)$ of the following function, simplifying your answer.

$$f(x) = \frac{\sin x}{\sqrt{x+1}}.$$

(7 marks)

(c) Determine the exact value of

$$\int_0^{\frac{\pi}{2}} \left(\frac{\cos x}{\sqrt{x+1}} - \frac{\sin x}{2\sqrt{(x+1)^3}} \right) dx\,.$$

(7 marks)

Question 4

(a) Interpret

$$\lim_{n\to\infty} \sum_{k=1}^{n} \frac{2k}{3n^2+k^2}\ ,$$

as a definite integral and evaluate it.

(6 marks)

(b) Show that the following reduction formula holds.

$$\int x^n e^x\, dx = x^n e^x - n \int x^{n-1} e^x\, dx\,.$$

(7 marks)

(c) Evaluate the following integral.

$$\int_0^1 x^3 e^x\, dx\ .$$

(7 marks)

Question 5

Evaluate the following integrals, showing your workings clearly.

(a)

$$\int_1^3 \frac{1}{e^x + e^{-x}}\, dx\,.$$

(10 marks)

(b)

$$\int_1^2 x(1-x)^{2022}\, dx\,.$$

(10 marks)

----- END OF PAPER -----

1(a) $\int (\sin x)^{2022} \cos x \, dx = \frac{1}{2023} \int (2023)(\cos x)(\sin x)^{2022} \, dx$

$= \frac{1}{2023} (\sin x)^{2023} + C$ where C is a constant

(b) $\int 3x \cos(1-2x) \, dx = -\frac{1}{2} \int (3x)(-2\cos(1-2x)) \, dx$

$= -\frac{1}{2}\left[3x \sin(1-2x) - \int 3\sin(1-2x) \, dx \right]$

$= -\frac{1}{2}\left[3x \sin(1-2x) + \frac{3}{2} \int -2\sin(1-2x) \, dx \right]$

$= -\frac{1}{2}\left[3x \sin(1-2x) + \frac{3}{2} \cos(1-2x) \right] + C$

$= -\frac{3}{2} x \sin(1-2x) - \frac{3}{4} \cos(1-2x) + C,$

where C is a constant

(c) $x^3 + 3x^2 + 2x = x(x^2+3x+2) = x(x+1)(x+2)$

let $\frac{x+3}{x^3+3x^2+2x} = \frac{A}{x} + \frac{B}{x+1} + \frac{C}{x+2}$

$= \frac{A(x+1)(x+2) + Bx(x+2) + Cx(x+1)}{x(x+1)(x+2)}$

$= \frac{(A+B+C)x^2 + (3A+2B+C)x + 2A}{x(x+1)(x+2)}$

$\Leftrightarrow A+B+C = 0, \quad 3A+2B+C = 1, \quad 2A = 3$

$(3A+2B+C) - (A+B+C) = 1 - 0$

$2A + B = 1$

$B = 1 - 2A = 1 - 3 = -2$

$A = \frac{3}{2}$

$C = -A - B = 2 - \frac{3}{2} = \frac{1}{2}$

$\therefore \int_1^2 \frac{x+3}{x^3+3x^2+2x} \, dx = \frac{1}{2} \int_1^2 \frac{3}{x} - \frac{4}{x+1} + \frac{1}{x+2} \, dx$

$= \frac{1}{2}\left[3\ln x - 4\ln(x+1) + \ln(x+2) \right]_1^2$

$= -\frac{5}{2} \ln 3 + \frac{9}{2} \ln 2$

2(a) let $y_1 = 5+x$

let $y_2 = -x^2-3x+10$

$y_2 = -(x^2+3x-10)$

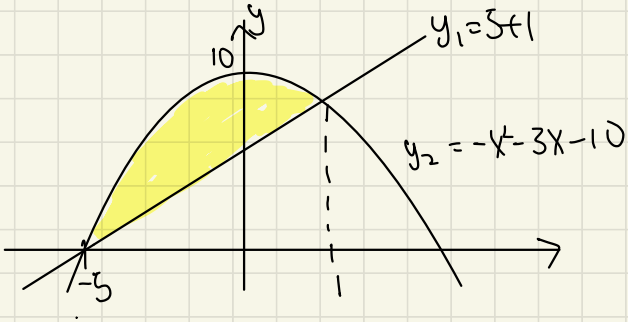
$= -(x+5)(x-2)$

let $y_1 = y_2$

$5+x = -x^2-3x+10$

$x^2+4x-5 = 0$

$(x+5)(x-1) = 0$



area of region R

$= \int_{-5}^{1} y_2 - y_1 \, dx$

$= \int_{-5}^{1} 5-2x-x^2 \, dx$

$= \left[5x - x^2 - \frac{1}{3}x^3\right]_{-5}^{1}$

$= 12$

(b) required volume

$= \int_{-5}^{1} \pi y_2^2 - \pi y_1^2 \, dx$

$= \pi \int_{-5}^{1} y_2^2 - y_1^2 \, dx$

$= \pi \int_{-5}^{1} x^4 + 6x^3 - 12x^2 - 50x + 75 \, dx$

$= \pi \left[\frac{1}{5}x^5 + \frac{3}{2}x^4 - 4x^3 - 25x^2 + 75x\right]_{-1}^{5}$

$= \frac{1176}{5}\pi$

(c) required volume

$= \int_{-1}^{5} 2\pi r(x) h(x) \, dx$ where $r(x) = 3-x$, $h(x) = y_2 - y_1 = 5-4x-x^2$

$= 2\pi \int_{-1}^{5} (3-x)(5-4x-x^2) \, dx$

$= 2\pi \int_{-1}^{5} x^3 + x^2 - 17x + 15 \, dx$

$= 2\pi \left[\frac{x^4}{4} + \frac{x^3}{3} - \frac{17}{2}x^2 + 15x\right]_{-1}^{5}$

$= 168\pi$

3(a) let $g(t) = \cos(\cos(t))$

$$\int_{e^x}^{x^2} \cos(\cos(t))\,dt = G(x^2) - G(e^x)$$

$$\frac{d}{dx}\int_{e^x}^{x^2} \cos(\cos(t))\,dt = \frac{d}{dx}\left[G(x^2) - G(e^x)\right]$$

$$= \frac{d}{dx}\left[G(t_1) - G(t_2)\right] \text{ where } t_1 = x^2,\ t_2 = e^x$$

$$= \frac{dG}{dt_1}\cdot\frac{dt_1}{dx} - \frac{dG}{dt_2}\cdot\frac{dt_2}{dx}$$

$$= g(t_1)\frac{dt_1}{dx} - g(t_2)\frac{dt_2}{dx}$$

$$= g(x^2)(2x) - g(e^x)(e^x)$$

$$= 2x\cos(\cos(x^2)) - e^x\cos(\cos(e^x))$$

(b) $f(x) = (x+1)^{-\frac{1}{2}}\sin(x)$

$$f'(x) = (x+1)^{-\frac{1}{2}}\cos(x) + \left(-\tfrac{1}{2}\right)(x+1)^{-\frac{3}{2}}\sin x$$

$$= \frac{\cos x}{(x+1)^{\frac{1}{2}}} - \frac{\sin x}{2(x+1)^{\frac{3}{2}}}$$

(c) $$\int_0^{\frac{\pi}{2}} \frac{\cos x}{\sqrt{x+1}} - \frac{\sin x}{2\sqrt{(x+1)^3}}\,dx = \left[\frac{\sin x}{\sqrt{x+1}}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{\sqrt{1+\frac{\pi}{2}}}$$

$$= \frac{2}{\sqrt{4+2\pi}}$$

4(a) $\lim_{n\to\infty} \sum_{k=1}^{n} \frac{2k}{3n^2+k^2} = \lim_{n\to\infty} \underbrace{\frac{1}{n}}_{\Delta x} \cdot \underbrace{\frac{2k}{3n+\frac{k^2}{n}}}_{}$

$= \lim_{n\to\infty} \underbrace{\frac{1}{n}}_{\Delta x} \cdot \underbrace{\frac{2(\frac{k}{n})}{3+(\frac{k}{n})^2}}_{}$

$x_k = x_0 + k\Delta x = 0 + (\frac{k}{n}) = \frac{k}{n}$

$f(x_k) = f(\frac{k}{n})$

$\Rightarrow f(x) = \frac{2x}{3+x^2}$

$= \int_0^1 \frac{2x}{3+x^2}\,dx$

$= \left[\ln(x^2+3)\right]_0^1$

$= \ln 4 - \ln 3$

(b) using integration by parts

let $u = x^n$, $dv = e^x\,dx$

$du = n x^{n-1}\,dx$ $\quad v = e^x$

$\int x^n e^x\,dx = \int u\,dv$

$= uv - \int v\,du$

$= x^n e^x - \int e^x (n x^{n-1})\,dx$

$= x^n e^x - n\int x^{n-1} e^x\,dx$

(c) $\int x^3 e^x\,dx = x^3 e^x - 3\int x^2 e^x\,dx$

$= x^3 e^x - 3[x^2 e^x - 2\int x e^x\,dx]$

$= x^3 e^x - 3[x^2 e^x - 2[x e^x - \int e^x\,dx]] + C$

$= x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C$

$= e^x(x^3 - 3x^2 + 6x - 6) + C,$

where C is a constant

5(a) let $u(x) = e^x$

$du = e^x\,dx$

$$\int \frac{1}{e^x+e^{-x}}\,dx = \int \frac{e^x}{e^{2x}+1}\,dx$$

$$= \int \frac{1}{u^2+1}\,du$$

$= \tan^{-1} u + C$, where C is a constant

$$\int_1^3 \frac{1}{e^x+e^{-x}}\,dx = \left[\tan^{-1} u\right]_{u(1)}^{u(3)}$$

$$= \left[\tan^{-1} u\right]_{e}^{e^3}$$

$$= \tan^{-1} e^3 - \tan^{-1} e$$

(b) let $u(x) = 1-x$ $\quad x = 1-u$

$du = -\,dx$

$$\int x(1-x)^{2022}\,dx = \int (1-u)u^{2022}(-1)\,du$$

$$= \int (u-1)u^{2022}\,du$$

$$= \int u^{2023} - u^{2022}\,du$$

$= \frac{1}{2024}u^{2024} - \frac{1}{2023}u^{2023} + C$, where C is a constant

$$\int_1^2 x(1-x)^{2022}\,dx = \left[\frac{1}{2024}u^{2024} - \frac{1}{2023}\right]_{u(1)}^{u(2)}$$

$$= \left[\frac{1}{2024}u^{2024} - \frac{1}{2023}u^{2023}\right]_0^{-1}$$

$$= \frac{4047}{4044552}$$